

A beginner's bioinformatics guide for epi-seq analysis

Ming Tommy Tang

Youtube channel: Chatomics

[Divingintogeneticsandgenomics.com](https://divingintogeneticsandgenomics.com)

05/02/2025

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Lead the Bioinformatics effort for
**Cancer Immunologic Data
Commons in Shirley Liu lab**



AstraZeneca

Director of
Bioinformatics

2021- 2024.08.

Director of computational biology

Top 75 Bioinformatics
Blogs by feedspot.com



Dana-Farber
Cancer Institute

2020 – 2021
**Lead
Scientist**



Instructor of
Data Carpentries



2018-2020

**Senior
bioinformatician**

THE UNIVERSITY OF TEXAS
**MD Anderson
Cancer Center**
"Making Cancer History"

2014-2018

**Postdoc & Research
Scientist**



2014

PhD. Genetics & Genomics

Outstanding
International
student



2008 BS Biotechnology

Over 40 publications; > 5000 citations

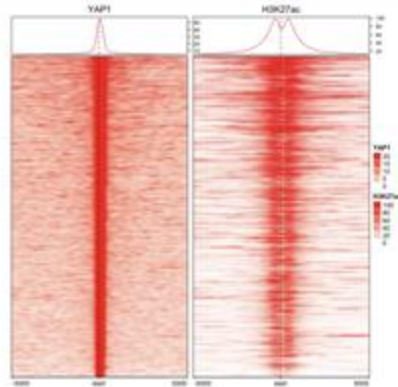
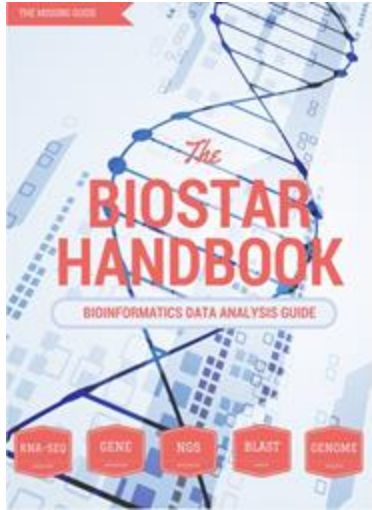
DNA CONFESSES DATA SPEAK

Blog: <https://divingintogeneticsandgenomics.com>
30K views per month



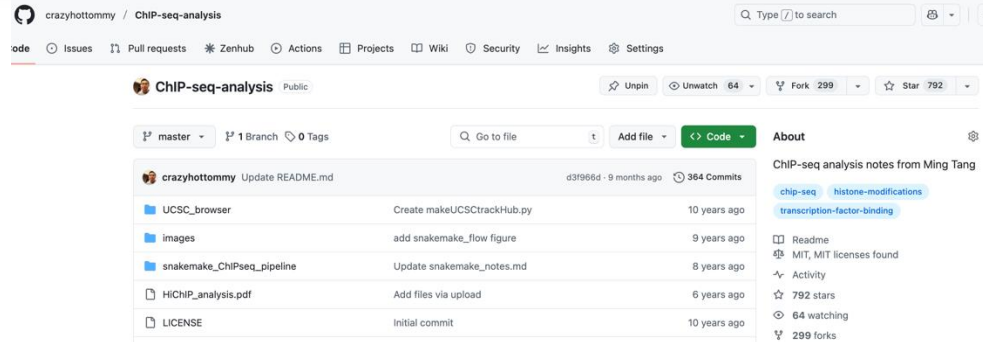
@tangming2005 ~38K followers
LinkedIn ~35K followers
Youtube: chatomics ~9000
subscribers

Download a step by step guide for free



ChIP-seq analysis Book chapter for
Biostar Handbook

Ming Tang

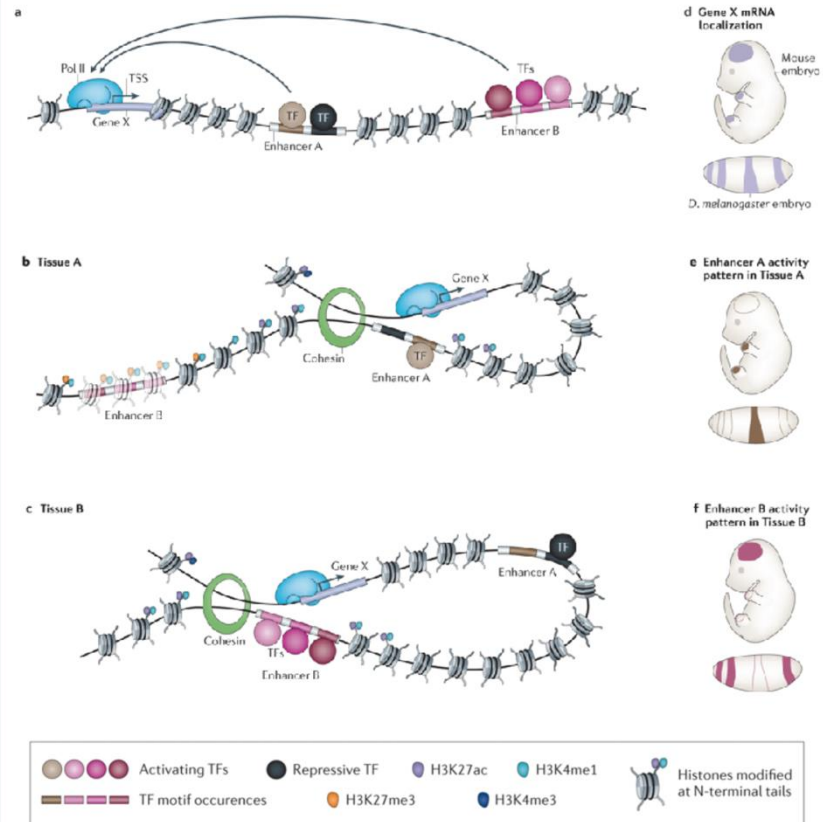


<https://divingintogeneticsandgenomics.com/publication/2017-08-01-biostarhandbook/>
<https://github.com/crazyhottommy/ChIP-seq-analysis>

Complexity in gene regulation

Diverse mechanisms to ensure that genes are expressed at the right time, in appropriate tissues and under specific conditions

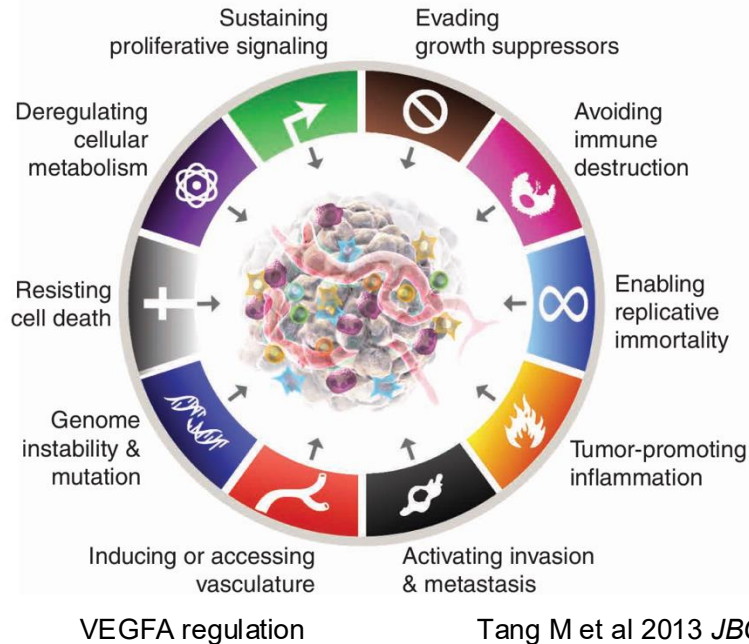
Numerous diseases associated with mutations in the non-coding genome



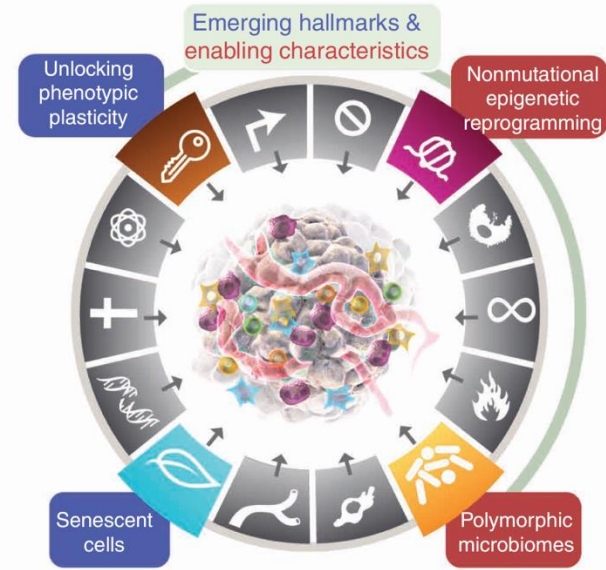
Nature Reviews | Genetics

Shlyueva, et al (2014). Transcriptional enhancers: from properties to genome-wide predictions.

Hallmarks of cancer: New Dimensions



Tang M et al 2013 *JBC*



Hypoxia: EMT

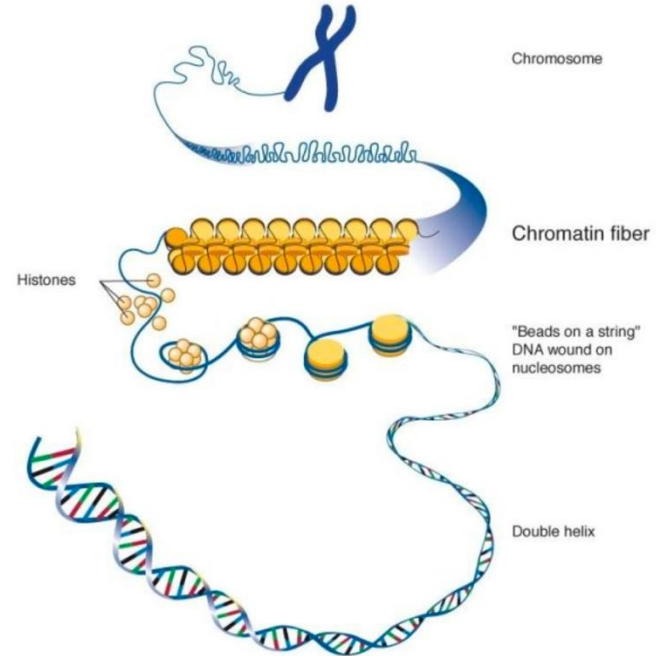
Tang M et al 2011, *PNAS*

Douglas Hanahan and Robert Weinberg, 2022, cancer discovery

2000, 2011, 2022

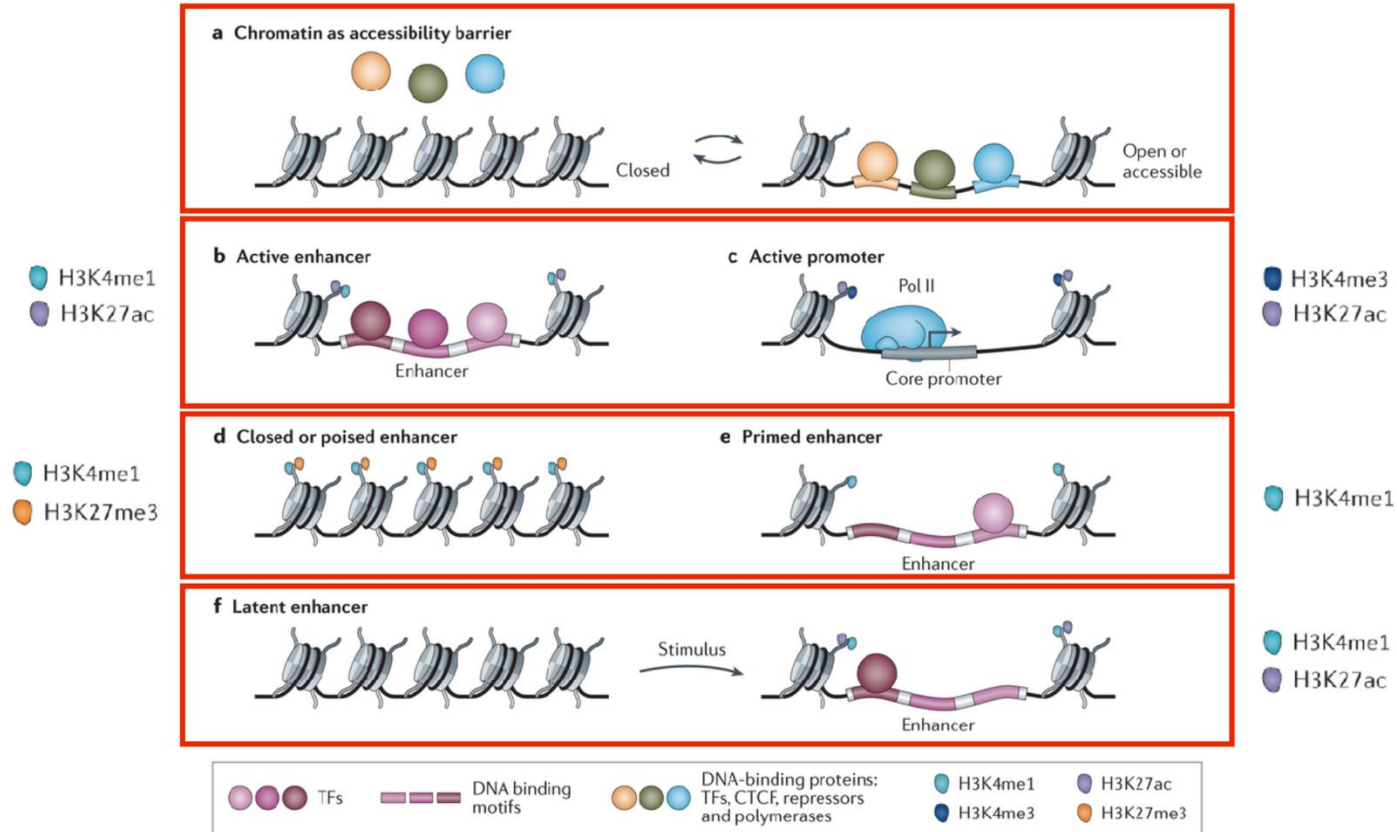
What is chromatin?

- **Chromatin:** a mixture of DNA and proteins that form the chromosomes found in the cells of humans and other higher organisms
- **Nucleosome:** 147 bp of DNA wound around 8 histone proteins (octamer) consisting of 2 copies each of the core histones (H2A, H2B, H3, H4)
- **Heterochromatin:** condensed chromatin
- **Euchromatin:** extended chromatin



<https://www.creative-diagnostics.com/blog/index.php/the-structure-and-function-of-chromatin/>

Histone Code



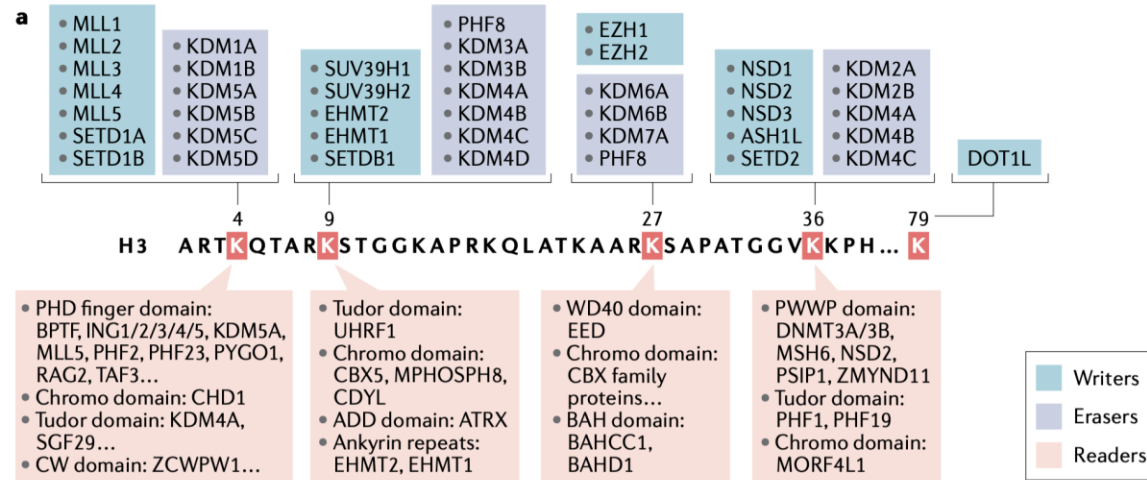
Shannan Ho Sui

Nature Reviews | Genetics

Shlyueva, et al (2014)

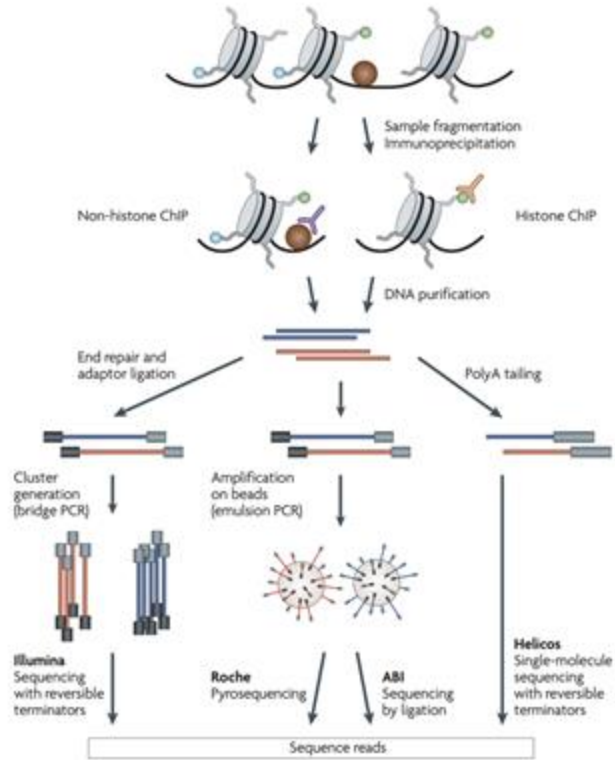
Chromatin structure determines if a gene is expressed or not

Balance between activities of readers and writers



HAT <-> HDAC

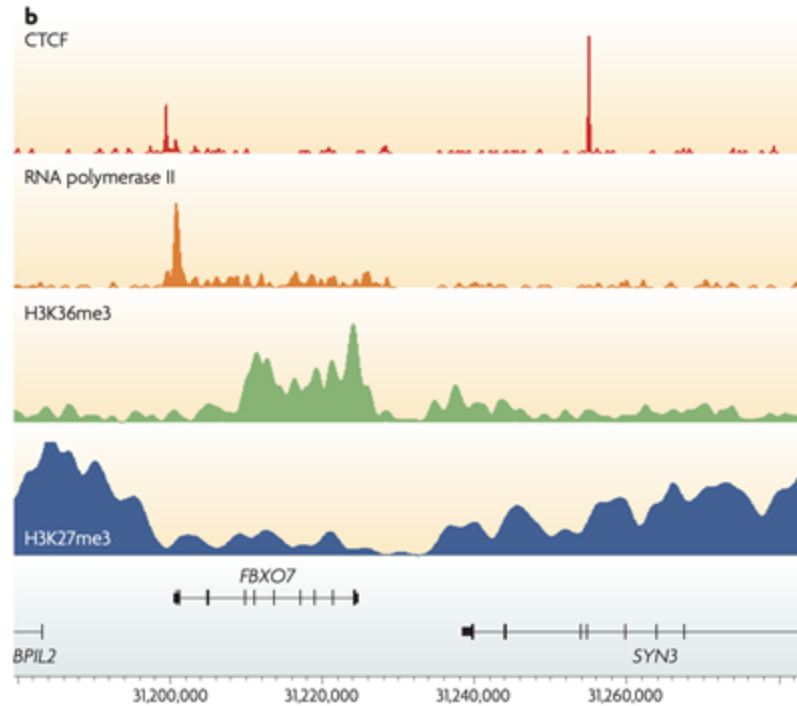
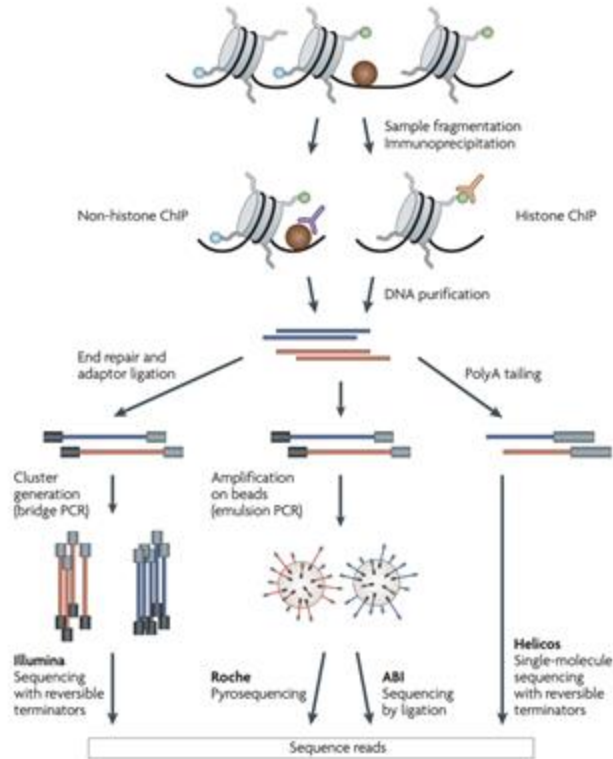
What is ChIP-seq



Peter Park <https://www.nature.com/articles/nrg2641>

<https://github.com/crazyhottommy/ChIP-seq-analysis>

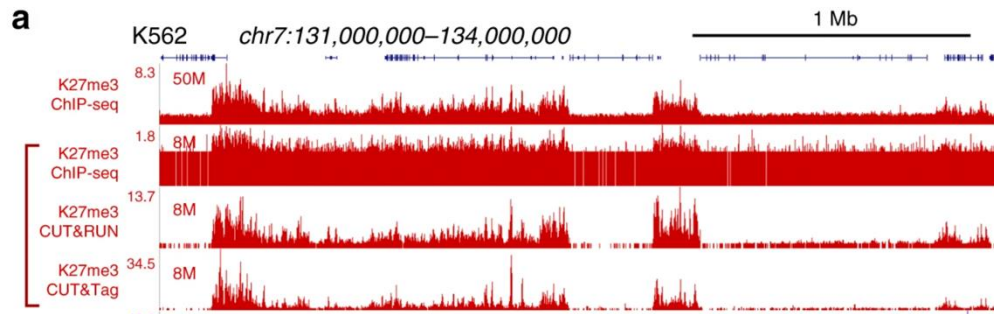
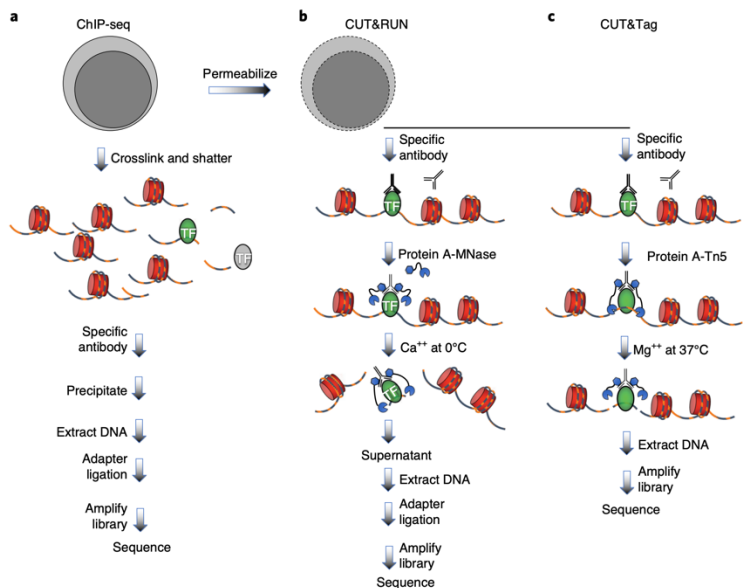
What is ChIP-seq



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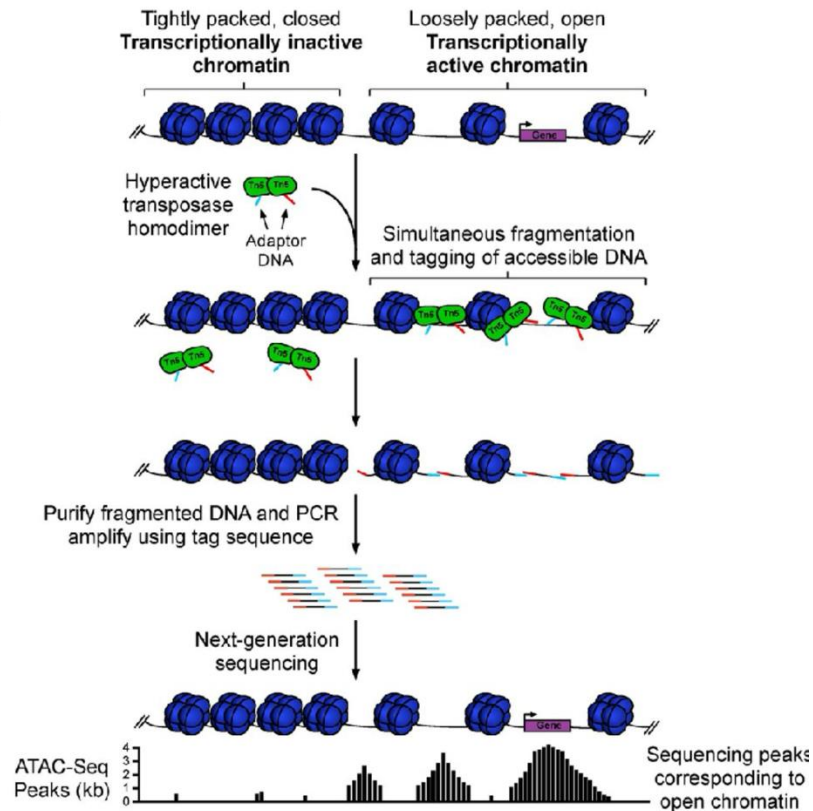
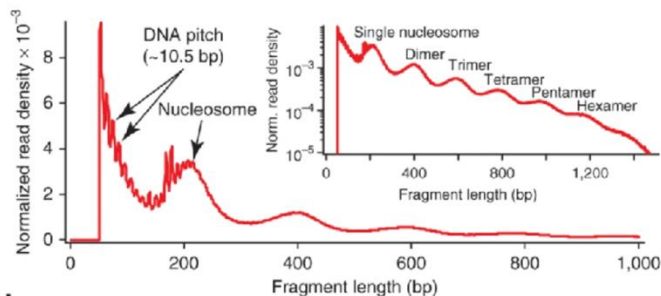
More recent development of epigenome profiling technologies: CUT&RUN CUT&Tag



Sequencing depth

Assay for Transposase-Accessible Chromatin with sequencing (ATAC-seq)

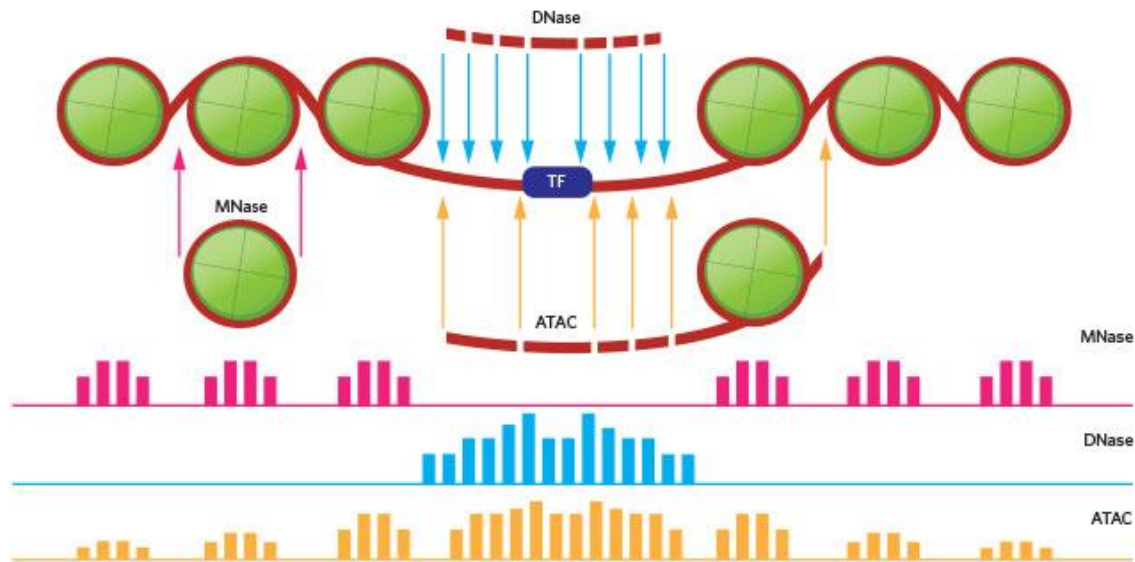
- Measure the extent to which DNA is open and accessible genome-wide
- Uses a hyperactive Tn5 transposase that cuts and inserts sequencing adapters into regions of chromatin that are accessible
- Fragment length correlates with nucleosome-free regions (less than 147bp) and mono-, di- and trinucleosome regions



<https://seandavi.github.io/AtacSeqWorkshop/articles/Workflow.html>

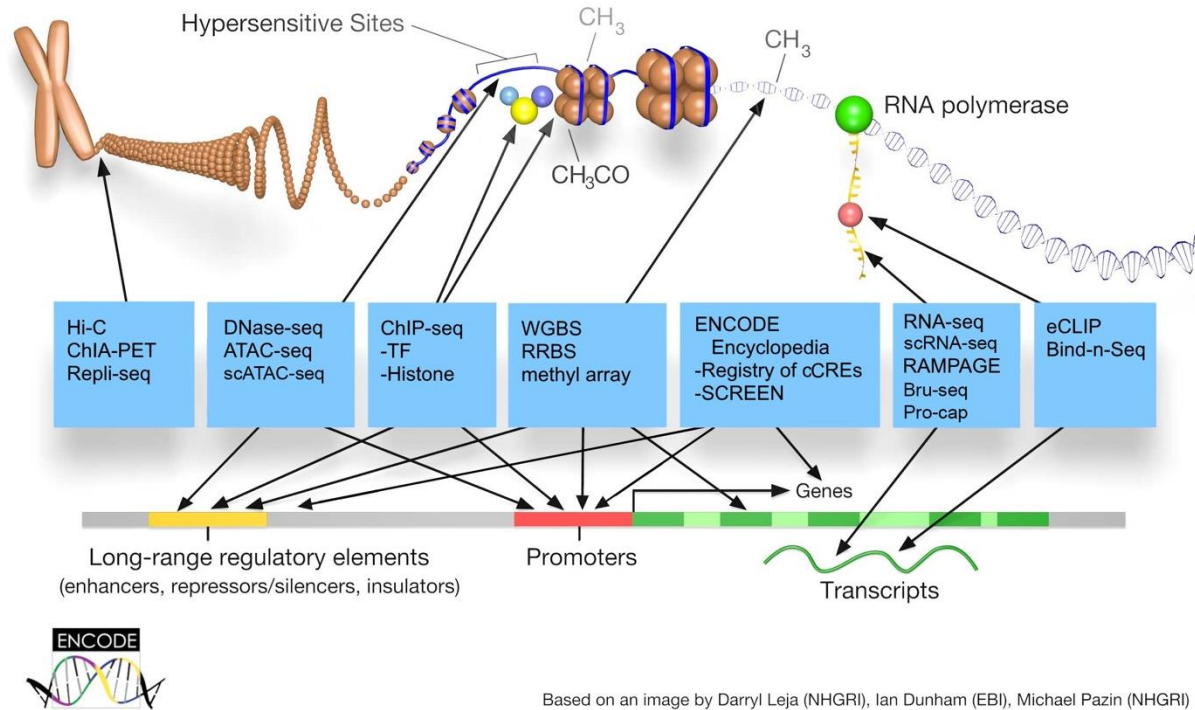
Shannan Ho Sui

ATACseq has gained popularity against MNase and Dnase-seq



<https://github.com/crazyhottommy/pyflow-ATACseq>

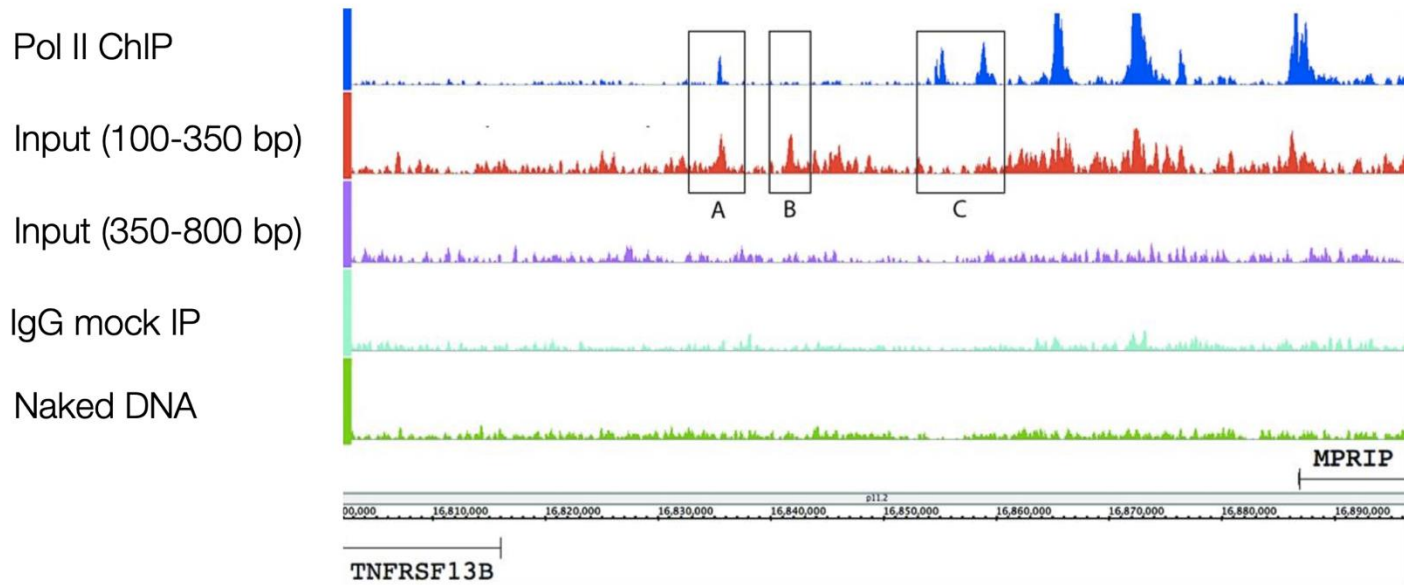
ENCODE: Encyclopedia of DNA Elements



Based on an image by Darryl Leja (NHGRI), Ian Dunham (EBI), Michael Pazin (NHGRI)

Controls

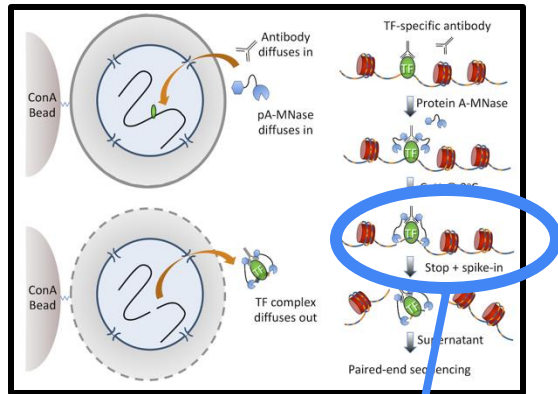
- ▶ ChIP-seq assays require input controls
- ▶ CUT&RUN uses a non-specific IgG control
 - ▶ Also recommend using a H3K4me3 positive control
- ▶ Controls are not typically used for ATAC-seq
 - ▶ Expensive and of limited value. A control for a given sample would be genomic DNA from the sample that, instead of transposase treatment, is fragmented (e.g. by sonication), has adapters ligated, and is sequenced along with the ATAC sample.



Map of ChIP-seq versus control signals

There are three spike-in methods that could be used for CUT&RUN protocols

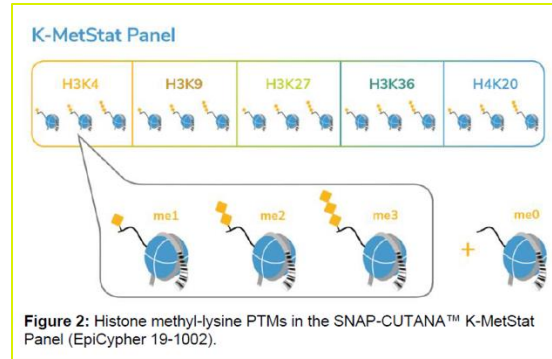
E. coli Spike-In



From Skene & Henikoff
(2017)

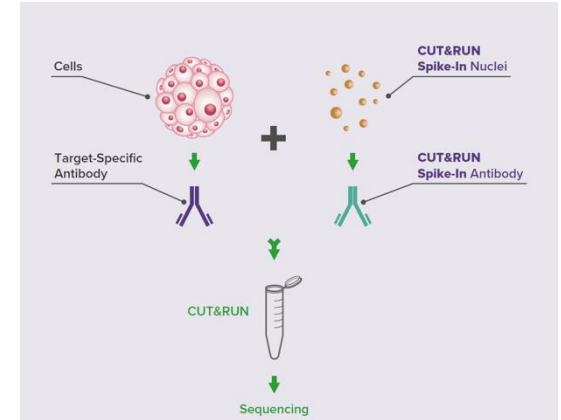
Sheared *E. coli* DNA
mixed with enzyme
inactivation buffer

K-met stat Spike-In



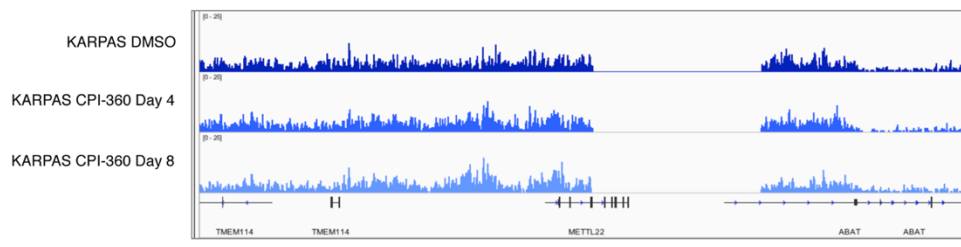
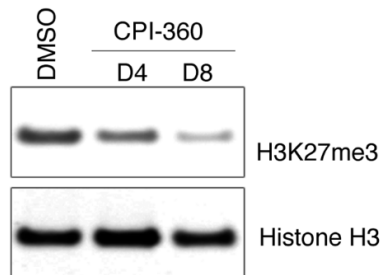
K-met stat panel of
recombinant nucleosomes
spiked into reaction prior to
antibody incubation

Drosophila Nuclei Spike-In



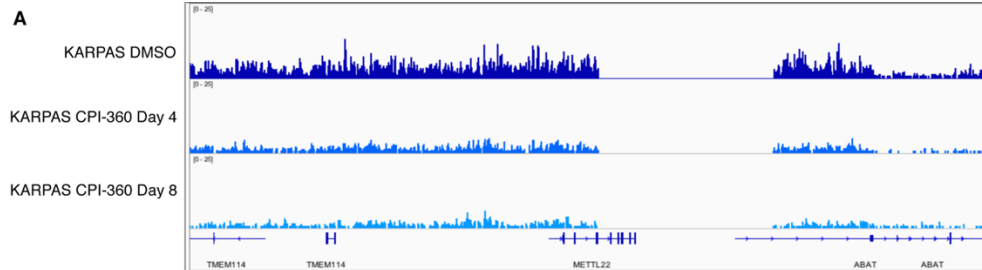
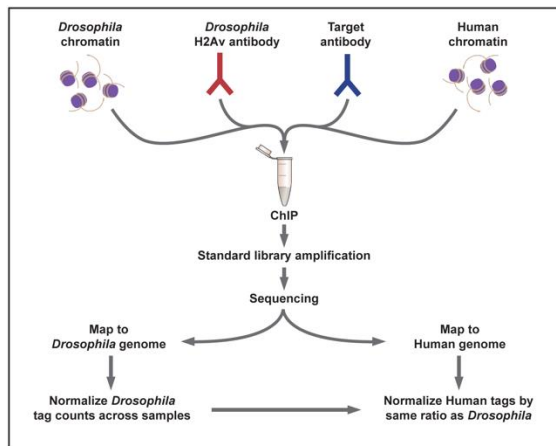
Cryopreserved *Drosophila*
nuclei and *Drosophila* H2Av-
specific antibody spiked into
reaction

In two experiments we ask what is best normalization approach using a classical example from ChIP-seq where absolute abundance of H3K27me3 decreases after EZH2 inhibition.



Original, non-normalized tracks

Inclusion of *Drosophila* spike-in reaction



Peaks after spike-in normalization

From Egan et al (2016)

Analysis steps for ChIP-seq



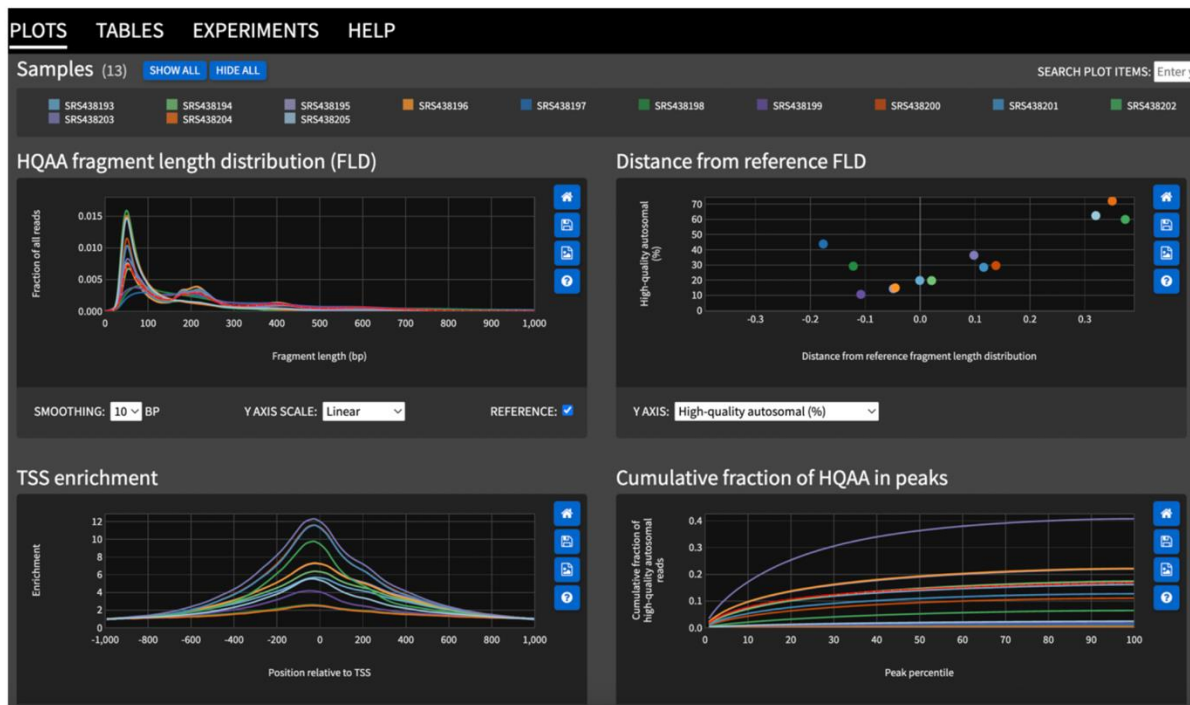
Fastq: raw sequencing file

Bam: aligned binary file, use
samtools → sam file

Bed: a text peak file with
chr, start, end

Bigwig: binary pileup file or
coverage file

Quality control with atacqv



Shannan Ho Sui

<https://divingintogeneticsandgenomics.com/post/calculate-scatacseq-tss-enrichment-score/>

Bigwig file

Building a signal map: BAM → BIGWIG

aim: **reduce the vast amount of information** from the BAM file to the simple information: **How many reads do I have** (per bp/genomic bin/...)?

BAM file

```
39V34V1:38:C0XLIACXX:4:1216:16137:31969 163 chr1 3000307 42 51M  
= 3000408 152  
CTGTAGTTACTGTTTGCCTACCTAGATTCTCTTTCCAGAATTCTCTTAG  
CCCCFFFFHHHHIIJJJJJJIIHHPGIGIJJJJJHIIHIIJJJJGOF AS:I:0  
XN:I:0 XM:I:0 XO:I:0 XG:I:0 NM:I:0 MD:Z:51 YS:I:0 YT:Z:CP
```



bedGraph/bigWig

chr2	100100	100120	5
chr2	100121	100141	3.2
chr2	100142	100163	13.8



Position (bp) 50 bp

the reduced size leads to many advantages of bigwig files over BAMs:

- data storage
- data sharing!
- intuitive visualization via genome browsers
- more efficient for down-stream analyses (...)

Bigwig file

Building a signal map: BAM → BIGWIG

aim: **reduce the vast amount of information** from the BAM file to the simple information: **How many reads do I have (per bp/genomic bin/...)?**

BAM file

```
39V34V1:38:CCXLIACXX:4:1216:16137:31969 163 chr1 3000307 42 51M
= 3000408 152
CTGTAGTTACTGTTTCTTACCTAGATTCTCTCTTCCAGATTCTCTTAG
CCCTTTTTTTHHHHIIJJJJJJJHGHFGIGIJJJJJHGHGIIIIJJGF AS:I:0
XN:I:0 XM:I:0 XD:I:0 XG:I:0 NM:I:0 MD:Z:51 YS:I:0 YT:Z:CP
```



bedGraph/bigWig

chr2	100100	100120	5
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chr2	100142	100163	13.8



- the reduced size leads to many advantages of bigwig files over BAMs:
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 - data sharing!
 - intuitive visualization via genome browsers
 - more efficient for down-stream analyses (...)

Friederike Dündar, MPI-IE, Freiburg

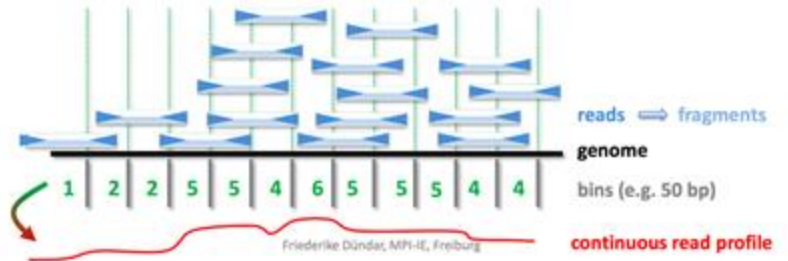
How coverage files are computed

experiment: DNA → sonicated to 200 bp fragments → 30-50 bp reads

given the reads, we try to retrace the steps of the experimental workflow!

bioinfo:

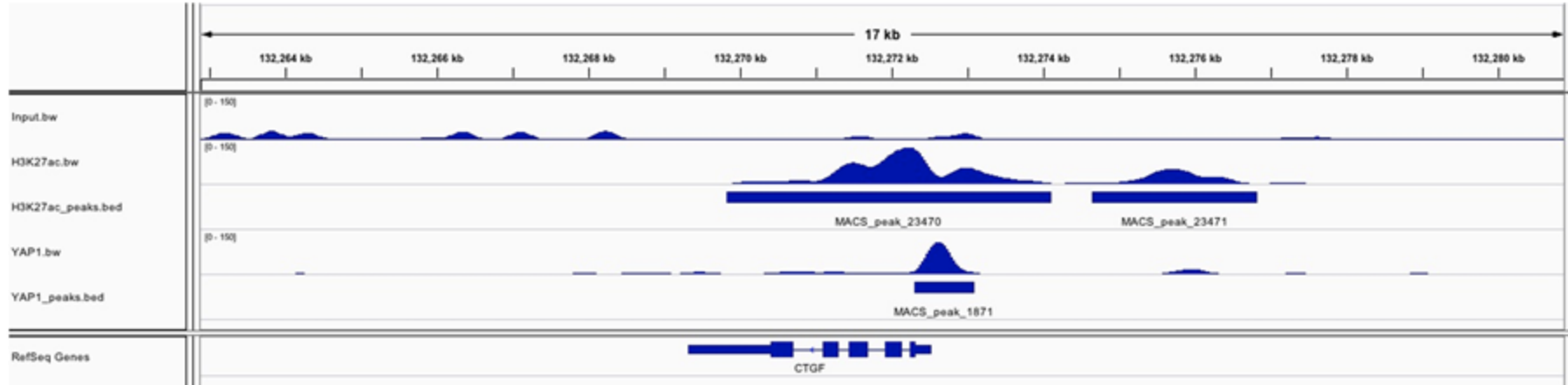
1. reads are extended to match the fragment size
2. the genome is binned, e.g. into windows of 50 bp
3. the number of extended reads (= "fragments") per genomic window is counted



Peak calling and visualize in IGV

MACS1/2, Homer

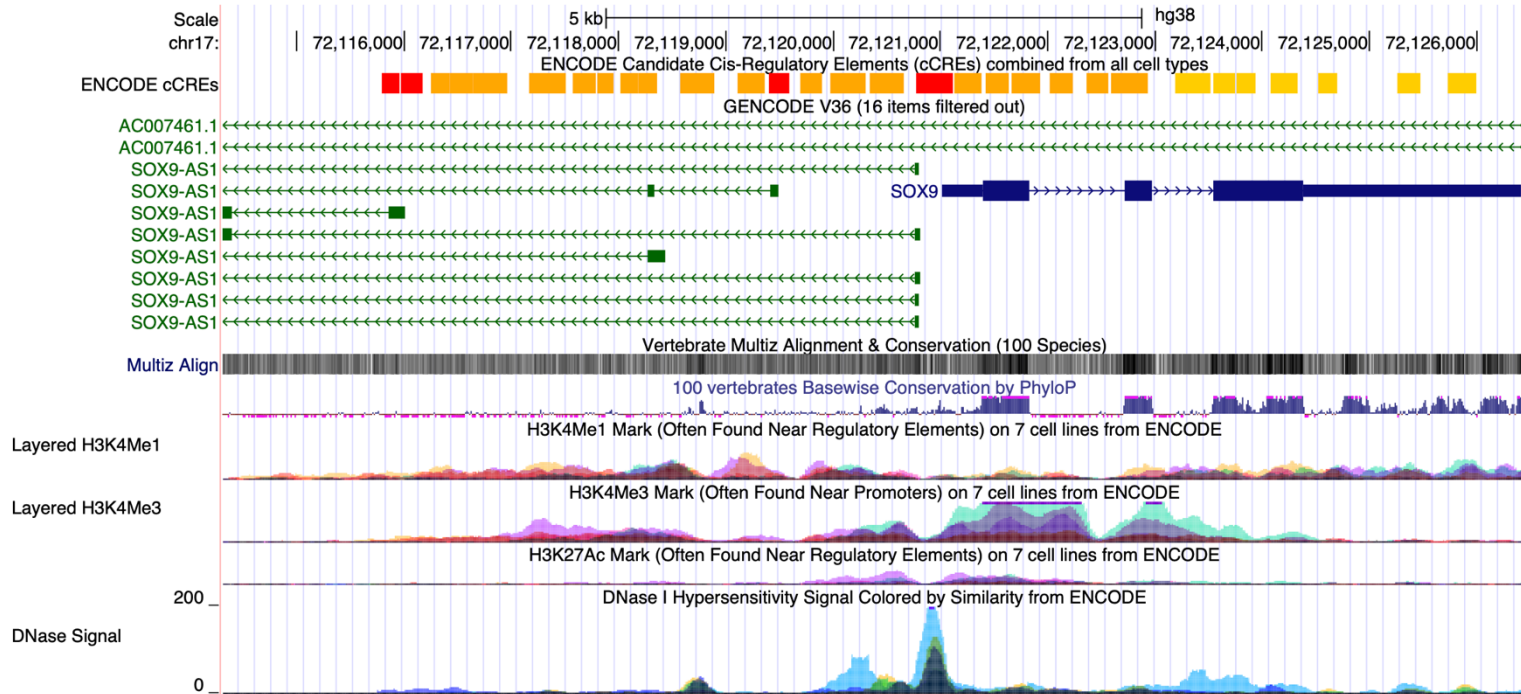
YAP1 binds to known target gene *CTGF*:



Best quality control with your eyes :)

Broad vs narrow peak

Visualizing peaks in the UCSC genome browser

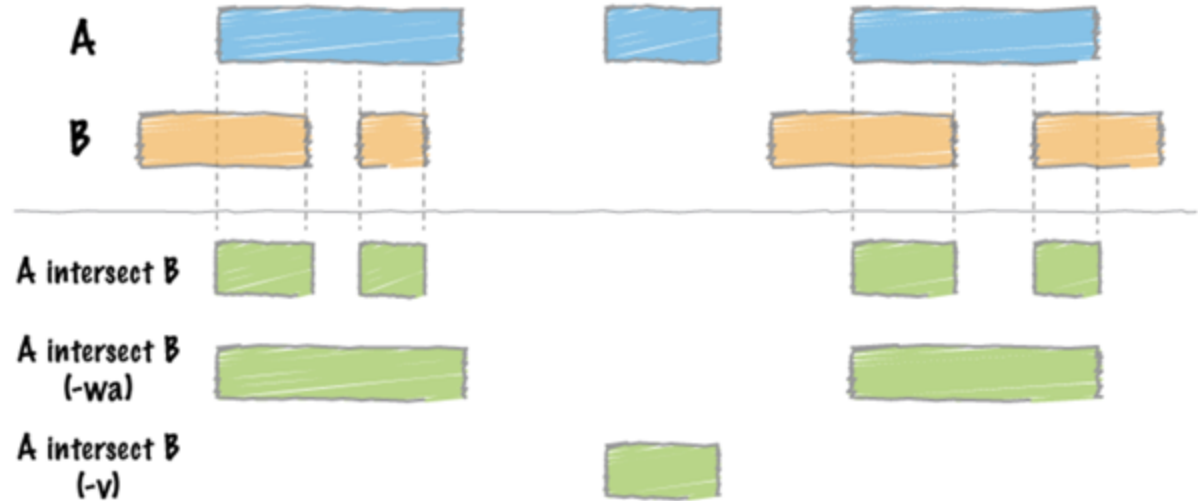


bedtools

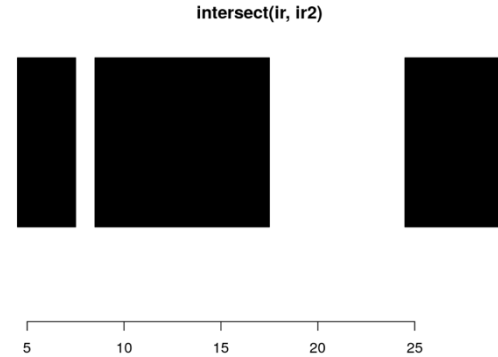
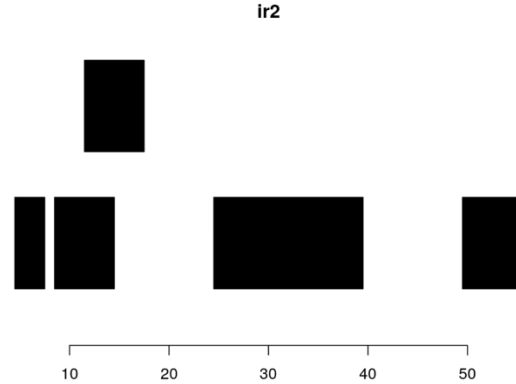
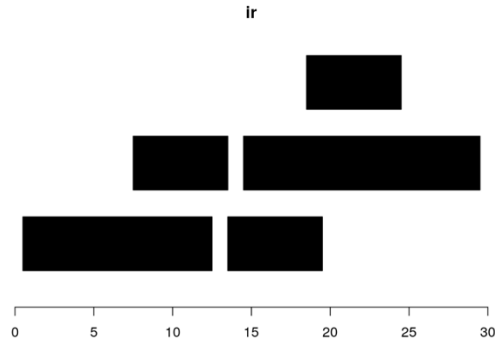
intersect



**Intersect w/
1 database**



GenomicRanges in R



<https://divingintogeneticsandgenomics.com/post/genomic-interval/>

Peak annotation
ChIPseeker

The end results ChIPseq, ATACseq, Cut&Run are count tables

RNAseq

	control1	control2	treat1	treat2
gene1	3	4	12	15
gene2	10	12	9	11
...				
geneN	45	39	10	12

Epi-seq

	control1	control2	treat1	treat2
peak1	3	4	12	15
peak2	10	12	9	11
...				
peakN	45	39	10	12

Differentially binding analysis

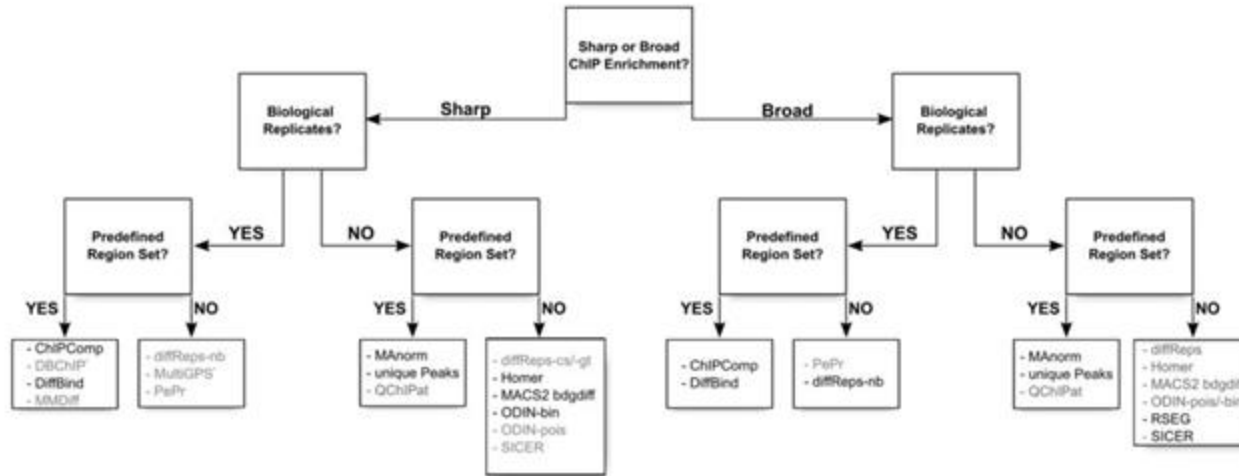
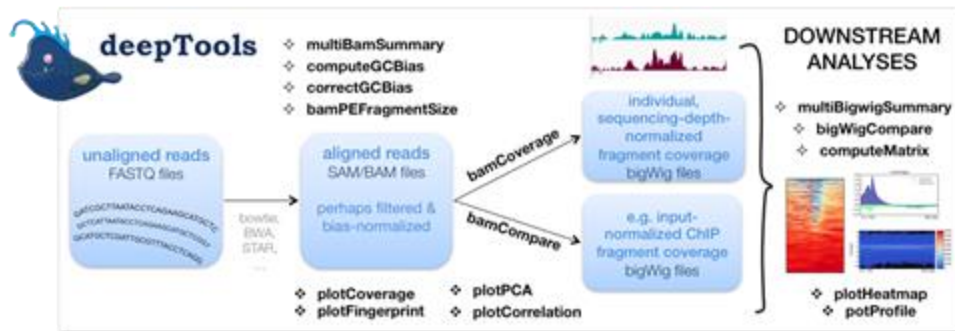
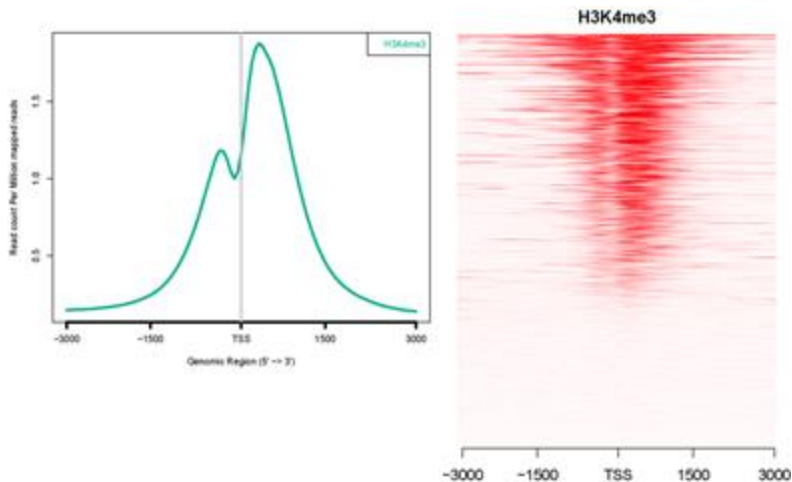


Figure 7. Decision tree indicating the proper choice of tool depending on the data set: shape of the signal (sharp peaks or broad enrichments), presence of replicates and presence of an external set of regions of interest. We have indicated in dark the name of the tools that give good results using default settings, and in gray the tools that would require parameter tuning to achieve optimal results: some tools suffer from an excessive number of DR (PePr, ODIN-pois), an insufficient number of DR (QChIPat, MMDiff, DBChIP) or from an imprecise definition of the DR for sharp signal (SICER, diffReps-nb). *MultiGPS has been explicitly developed for transcription factor ChIP-seq.

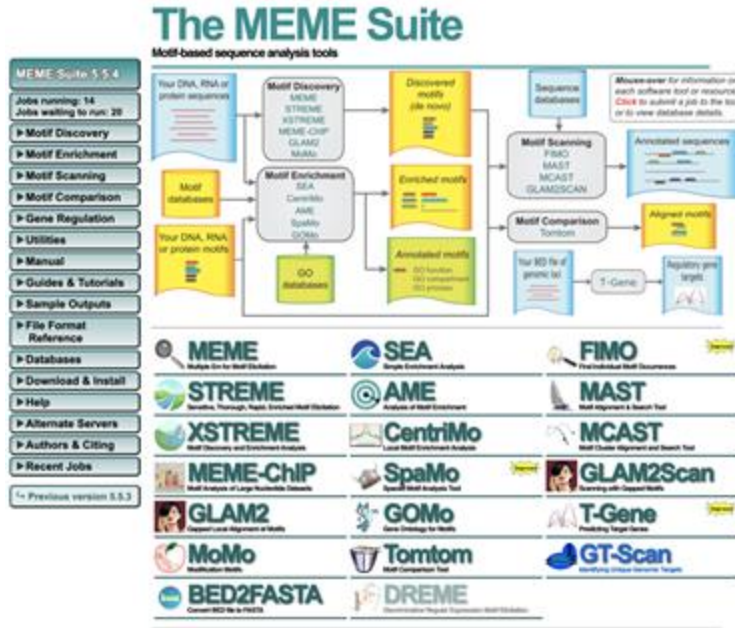
Meta plot and heatmap



<https://deeptools.readthedocs.io/en/latest/>

EnrichedHeatmap by Zuguang Gu

Motif analysis



Homer de novo Motif Results

Known Motif Enrichment Results
Gene Ontology Enrichment Results
If Homer is having trouble matching a motif to a known motif, try copy/pasting the matrix file into STAMP.
More information on motif finding results: HGMDB | Description of Results | Tips
Total target sequences = 37301
Total background sequences = 35962
* -- possible false positive

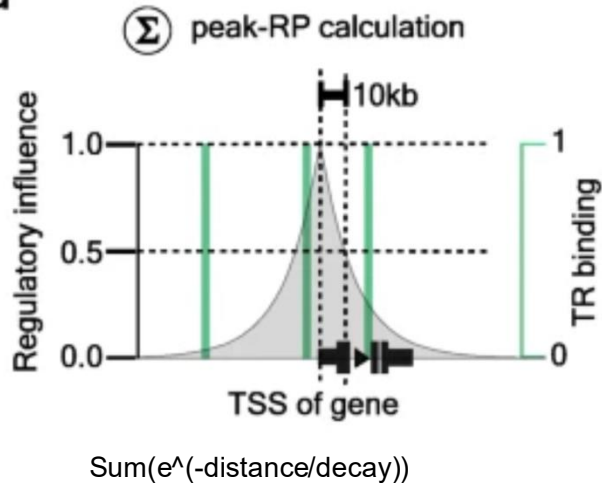
Rank/Motif	P-value	log P-value	% of Targets	% of Background	STDDBg/STD	Best Match/Details	Motif File
1 TGTTTACATA	1e-12661	-2.915e+04	70.91%	15.19%	40.5bp (65.1bp)	Foxa2/Forkhead/Liver-Foxa2-ChIP-Seq/Homer More Information Similar Motifs Found	motif file (fasta)
2 CTTGGCAG	1e-578	-1.332e+03	27.14%	16.52%	54.0bp (65.5bp)	NP1-haHmr/CTT/LNCaP-NP1-ChIP-Seq/Homer More Information Similar Motifs Found	motif file (fasta)
3 TTTTATTGGC	1e-384	-8.860e+02	17.77%	10.53%	53.9bp (62.1bp)	Unknown/Homocob/Lamb-p300-ChIP-Seq/Homer More Information Similar Motifs Found	motif file (fasta)
4 CTCTGTAAAT	1e-164	-3.783e+02	3.17%	1.28%	52.2bp (62.9bp)	PH048.1_Hwa13 More Information Similar Motifs Found	motif file (fasta)
5 ATGACTCA	1e-151	-3.485e+02	3.38%	1.47%	50.2bp (65.4bp)	NP1-232ZP/K562-NP1-232ZP-Seq/Homer More Information Similar Motifs Found	motif file (fasta)
6 CCCACTCTCTGG	1e-107	-2.485e+02	1.21%	0.35%	56.3bp (69.7bp)	CTCF/CD4+CTCF-ChIP-Seq/Homer More Information Similar Motifs Found	motif file (fasta)
7 AGATAAGATC	1e-72	-1.871e+02	2.10%	1.02%	55.1bp (58.5bp)	MA029.1_Ev1 More Information Similar Motifs Found	motif file (fasta)

Regulatory potential score

- We always want to integrate the epigenome sequencing data with RNAseq data. With Regulatory potential score per gene, it is very easy to integrate those two data types.
- A score of 0 means no binding for that gene. Ideally, you need chromatin interaction data to demonstrate that, but it is a close approx....
- When the data is ATACseq, the regulatory potential can be interpreted as gene activity score. (If we assume open chromatin = more active transcription)
- When the data is TF cut&run, the score can be interpreted as how likely a certain gene is regulated by that TF

How to calculate regulatory potential

a



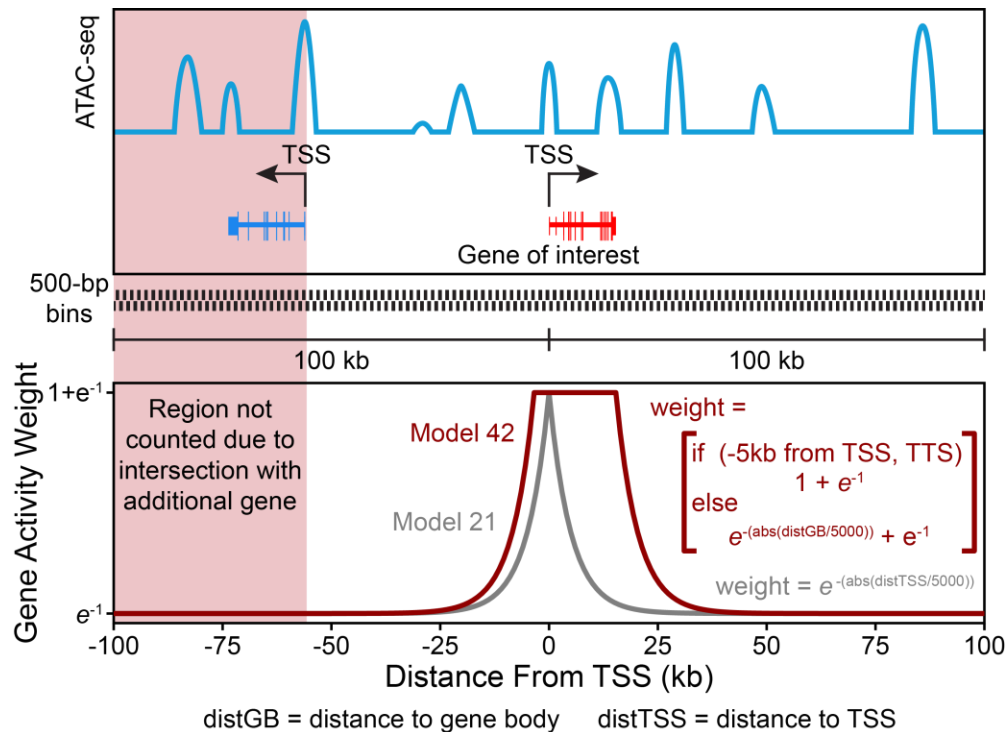
Consider that within 150kb upstream and downstream (default is decay * 15) of a gene, you have total 3 peaks.

- a peak at the TSS of a gene, the RP score is 1.
- a peak at -20 kb upstream, the RP score is $1 * e^{-(20/10)} = 0.25$, assuming the default decay rate is 10kb.
- a peak at +40kb downstream, the RP score is $1 * e^{-(40/10)} = 0.0625$.
The total RP score for that gene is $1 + 0.25 + 0.0625 = 1.3125$

The half-decay distance is 10kb if less than 20% of peaks overlap promoters

The half-decay distance is 1kb if more than 20% of peaks overlap promoters

Gene activity score calculating in ArchR (single-cell ATACseq analysis)



Further reading



Browse

Publish

 OPEN ACCESS

EDUCATION

Practical Guidelines for the Comprehensive Analysis of ChIP-seq Data

Timothy Bailey  , Pawel Krajewski , Istvan Ladunga , Celine Lefebvre , Qunhua Li , Tao Liu ,
Pedro Madrigal  , Cenny Taslim , Jie Zhang 

➤ [Genome Res.](#) 2012 Sep;22(9):1813–31. doi: 10.1101/gr.136184.111.

ChIP-seq guidelines and practices of the ENCODE and modENCODE consortia

What questions do you have?

Tip #1 Get on social medium

- Get on social medium: Twitter/X, Mastodon, bulesky, LinkedIn
- Follow people of the same interest; bioinformatics papers



I started using twitter after reading Stephen Turner's blog:
How to stay current on bioinformatics:
<https://www.r-bloggers.com/2017/02/staying-current-in-bioinformatics-genomics-2017-edition/>

Linkedin tips

- 1. take a professional picture
- 2. use a related banner



- 3. headline showing what's your expertise



Ming "Tommy" Tang
Cure Cancer with Data | Computational Biology | Data Science |
Biotech Executive | Open Science Advocate | Prev. Dana-Farber |
Harvard | MD Anderson | Join 16K followers on twitter
@tangming2005 to learn computation
Talks about #teamwork, #leadership, #management, ##medicine, and
#datascience

Tip #2 Write a blog

I web, therefore I am

Yihui Xie

Why a blog?

- a blog post is much better than a statement “good at R or Bayesian stats” on your CV
- “spend 30 minutes each day in 5 years building a website” vs “20 hours to write a CV in the last semester”
- there are many things that are more suitable for web pages (see my blog for example)

Credit: Yihui Xie

Why a blog? continued

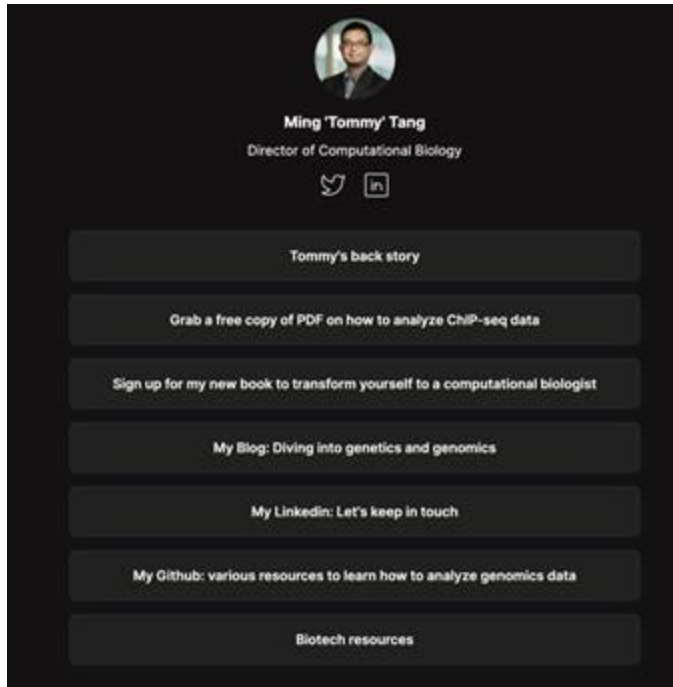
Paired Ends

Technical blogging for growth and learning

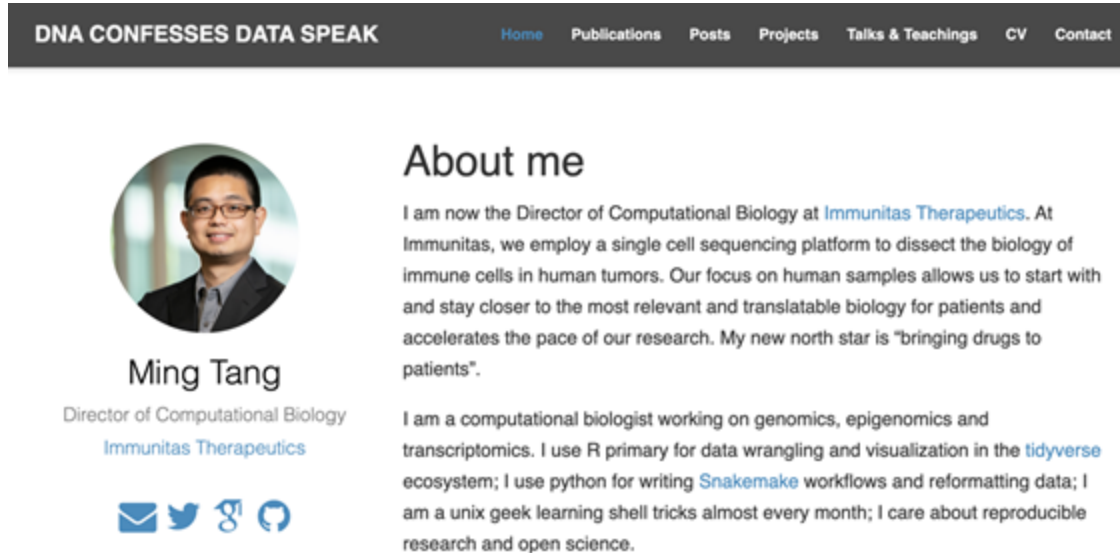
Writing a technical blog improves your writing, forces you to learn new things, helps others and yourself, and helps your career. This post: why blog, what to blog about, how to start, and using AI.

JAN 25, 2025

Build a website so others can find you



<https://tommytang.bio.link/>



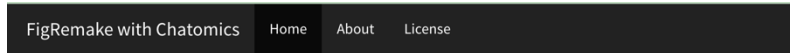
divingintogeneticsandgenomics.com

Start now

- If you do not have a website yet.
- The best time to start one is 10 years ago, the second best time is now.
- Take a weekend to set it up. Be visible: Blogdown or Quarto
- Make a github repository. Put your projects there.

More on Github

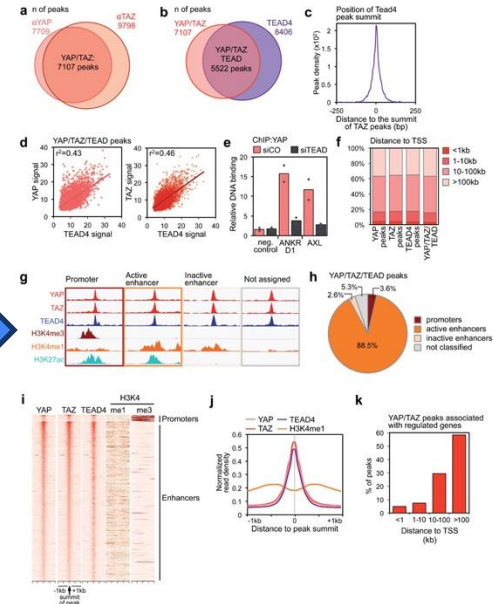
- The first thing I do as a hiring manager is to check if you have a github in your CV
- Put your course projects there if you are allowed
- Replicate published genomic paper figures.



Welcome! We are going to reproduce a figure from a genomics paper. Take a look at [About the project](#).

Step by step tutorial

1. Download fastq files.
2. Align fastq to genome
3. Generate bigwig files and call peaks
4. Re-create Fig 1 a,b,c
5. Re-create Fig 1 d,f
6. Re-create Fig 1 g,h
7. Re-create Fig 1 i,j,k
8. Now, it is your turn. [Accept the challenge](#)



https://crazyhottommy.github.io/reproduce_genomics_paper_figures/index.html

Tip #3 How to connect with people? on social media or in real life

One core thing to remember: always give value to the other side.

Be a giver not a taker

- Complement
- Ask questions. Be genuinely curious
- Offer help

Tip#4 What to consider in joining a company?

- 1. Are the science cutting edge?
- 2. Are the people you will work with kind and smart?
- 3. Is the finance of the biotech company good? Last at least 2-3 years

Credit: Thomas Tan

Make the transformation you want

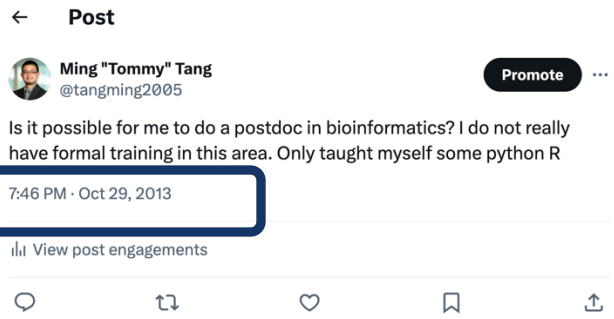


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Benefit

